

Week 3 – Assignment

Step 1: Setup Data –

Import the Superstore dataset into a table named store_raw. Create these 3 tables from it:

Customers

Orders

Products

Insert data into these tables using SELECT DISTINCT.

```
94  -- Verify customer Data
95
96 •  SELECT*
97  FROM customers
98  LIMIT 10;
99
100  -- =====
```

Result Grid	 Filter Rows:	 Export:	 Wrap Cell Content:	 Fetch rows:				
	customer_id	customer_name	segment	country	city	state	postal_code	region
▶	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West
	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South
	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South
	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West
	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central
	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central
	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West
	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West

```
120 -- Verify product Data
```

```
121
```

```
122 • SELECT *
```

```
123 FROM products;
```

```
124
```

```
125 -- =====
```

```
126
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
product_id	product_name	category	sub_category	
FUR-BO-10001798	Bush Somerset Collection Bookcase	Furniture	Bookcases	
FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs,...	Furniture	Chairs	
OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters b...	Office Supplies	Labels	
FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	Furniture	Tables	
OFF-ST-10000760	Eldon Fold 'N Roll Cart System	Office Supplies	Storage	
FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Acces...	Furniture	Furnishings	
OFF-AR-10002833	Newell 322	Office Supplies	Art	
TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	Technology	Phones	
OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by S...	Office Supplies	Binders	
OFF-AP-10002892	Belkin F5C206VTEL 6 Outlet Surge	Office Supplies	Appliances	
FUR-TA-10001539	Chromcraft Rectangular Conference Tables	Furniture	Tables	
TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	Technology	Phones	

products 5 x

```
158 -- Verify order Data
```

```
159
```

```
160 • SELECT *
```

```
161 FROM orders;
```

```
162
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0	41.9136
CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	OFF-LA-10000240	14.62	2	0	6.8714
US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	OFF-ST-10000760	22.368	2	0.2	2.5164
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	FUR-FU-10001487	48.86	7	0	14.1694
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	OFF-AR-10002833	7.28	4	0	1.9656
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2	90.7152
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	OFF-BI-10003910	18.504	3	0.2	5.7825
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	OFF-AP-10002892	114.9	5	0	34.47
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2	90.7152

orders 6 x

Step 2: Perform Required Queries—

1. Find all orders where sales are greater than the average sales. (Subquery)

```
172 • SELECT *
173 FROM orders
174 WHERE CAST(sales AS DECIMAL(10,2))
175 >
176 (
177     SELECT AVG(CAST(sales AS DECIMAL(10,2)))
178     FROM orders
179 )
180
```

order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0	41.9136
CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2	90.7152
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002033	911.424	4	0.2	68.3568
CA-2016-161389	12-05-2016	12-10-2016	Standard Class	IM-15070	OFF-BI-10003656	407.976	3	0.2	132.5922
CA-2014-105893	11-11-2014	11/18/2014	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0	13.3176
CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0	240.2649
US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5	-1665.0522
CA-2016-117590	12-08-2016	12-10-2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2	123.4737
CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32	-46.9764

2. Find the highest sales order for each customer. (Subquery)

```
186 • SELECT *
187 FROM orders o
188 WHERE CAST(o.sales AS DECIMAL(10,2))
189 =
190 (
191     SELECT MAX(CAST(i.sales AS DECIMAL(10,2)))
192     FROM orders i
193     WHERE i.customer_id = o.customer_id
194 )
195 LIMIT 100;
196
```

order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0	240.2649
US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5	-1665.0522
CA-2016-117590	12-08-2016	12-10-2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2	123.4737
CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32	-46.9764
CA-2016-105816	12-11-2016	12/17/2016	Standard Class	JM-15265	TEC-PH-10002447	1029.95	5	0	298.6855
CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	FUR-CH-10003968	319.41	5	0.1	7.098
CA-2014-106376	12-05-2014	12-10-2014	Standard Class	BS-11590	OFF-AR-10002671	1113.024	8	0.2	111.3024
CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	TEC-AC-10000844	339.96	5	0.2	67.992
CA-2015-110711	06-07-2015	06-12-2015	Standard Class	UA-14020	OFF-ST-10003656	671.02	7	0	20.1570

3. Calculate total sales for each customer. (CTE)

```
201
202 • WITH customer_sales AS
203 (
204     SELECT
205         customer_id,
206         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
207     FROM orders
208     GROUP BY customer_id
209 )
210
211 SELECT *
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	customer_id	total_sales
▶	CG-12520	1148.78
	DV-13045	1119.48
	SO-20335	2602.58
	BH-11710	6255.34
	AA-10480	1790.51
	IM-15070	4930.49
	HP-14815	1990.31
	PK-19075	8646.93
	AG-10270	2582.90
	ZD-21925	1493.94
	KB-16585	2744.74
	CF-20025	762.54

Result 9 x

4. Find customers whose total sales are above average. (CTE + Subquery)

```
219 • WITH customer_sales AS
220 (
221     SELECT
222         customer_id,
223         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
224     FROM orders
225     GROUP BY customer_id
226 )
227
228 SELECT *
229 FROM customer_sales
230 WHERE total_sales >
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	customer_id	total_sales
▶	BH-11710	6255.34
	IM-15070	4930.49
	PK-19075	8646.93
	TB-21520	4737.48
	MA-17560	4299.16
	SN-20710	3127.97
	LC-16930	4492.94
	RA-19885	3832.32
	ES-14080	4657.93
	KM-16720	4909.47

Result 10 x

5. Rank all customers based on total sales. (Window Function)

```
241 • WITH customer_sales AS
242 (
243     SELECT
244         customer_id,
245         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
246     FROM orders
247     GROUP BY customer_id
248 )
249
250 SELECT
251     customer_id,
252     total_sales,
253     RANK() OVER (ORDER BY total_sales DESC) AS sales_rank
254 FROM customer_sales;
255
```

Result Grid

customer_id	total_sales	sales_rank
SM-20320	25043.07	1
TC-20980	19052.22	2
RB-19360	15117.35	3
TA-21385	14595.62	4
AB-10105	14473.57	5
KL-16645	14175.23	6
SC-20095	14142.34	7
HL-15040	12873.30	8

Result 11

6. Assign row numbers to each order within a customer. (Window Function + PARTITION BY)

```
256 -- =====
257 -- QUERY 6: WINDOW FUNCTION + PARTITION BY
258 -- Assign row numbers to orders within each customer
259 -- =====
260
261 • SELECT *,
262     ROW_NUMBER() OVER
263     (
264         PARTITION BY customer_id
265         ORDER BY CAST(sales AS DECIMAL(10,2)) DESC
266     ) AS row_num
267 FROM orders;
268
269 -- =====
270 -- QUERY 7: WINDOW FUNCTION
```

Result Grid

order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit	row_num
CA-2016-103982	03-03-2016	03-08-2016	Standard Class	AA-10315	OFF-SU-10000151	3930.072	3	0.2	-786.0144	1
CA-2014-128055	3/31/2014	04-05-2014	Standard Class	AA-10315	OFF-BI-10004390	673.568	2	0.2	252.588	2
CA-2016-103982	03-03-2016	03-08-2016	Standard Class	AA-10315	TEC-PH-10000895	431.976	3	0.2	32.3982	3
CA-2017-147039	6/29/2017	07-04-2017	Standard Class	AA-10315	OFF-AP-10000576	362.94	3	0	90.735	4
CA-2014-128055	3/31/2014	04-05-2014	Standard Class	AA-10315	OFF-AP-10002765	52.98	2	0	14.8344	5
CA-2016-103982	03-03-2016	03-08-2016	Standard Class	AA-10315	TEC-AC-10002857	41.72	7	0.2	5.7365	6
CA-2015-121391	10-04-2015	10-07-2015	First Class	AA-10315	OFF-ST-10001590	26.96	2	0	7.0096	7
CA-2014-138100	9/15/2014	9/20/2014	Standard Class	AA-10315	OFF-PA-10000349	14.94	3	0	7.0218	8

Result 12

7. Display top 3 customers based on total sales. (Window Function)

```
274 • WITH customer_sales AS
275 (
276     SELECT
277         customer_id,
278         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
279     FROM orders
280     GROUP BY customer_id
281 )
282
283 SELECT *
284 FROM
285 (
286     SELECT *,
287         RANK() OVER(ORDER BY total_sales DESC) AS sales_rank
288     FROM customer_sales
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	total_sales	sales_rank
▶	SM-20320	25043.07	1
	TC-20980	19052.22	2
	RB-19360	15117.35	3

Final Combined Query (JOIN + CTE + Window Function together)

```
298 • WITH customer_sales AS
299 (
300     SELECT
301         customer_id,
302         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
303     FROM orders
304     GROUP BY customer_id
305 )
306
307 SELECT
308     c.customer_name,
309     cs.total_sales,
310     RANK() OVER(ORDER BY cs.total_sales DESC) AS sales_rank
311 FROM customer_sales cs
312 JOIN customers c
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

	customer_name	total_sales	sales_rank
▶	Sean Miller	25043.07	1
	Sean Miller	25043.07	1
	Sean Miller	25043.07	1
	Sean Miller	25043.07	1
	Sean Miller	25043.07	1
	Tamara Chand	19052.22	6
	Tamara Chand	19052.22	6
	Tamara Chand	19052.22	6

Result 14 x

MINI PROJECT – CUSTOMER SALES INSIGHTS

1. Who are the top 5 customers?

```
322
323 • WITH customer_sales AS
324 (
325     SELECT
326         customer_id,
327         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
328     FROM orders
329     GROUP BY customer_id
330 )
331
332 SELECT *
333 FROM
334 (
335     SELECT *,
336         RANK() OVER(ORDER BY total_sales DESC) AS rnk
337     FROM customer_sales
338 ) x
339 WHERE rnk <= 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	total_sales	rnk
▶	SM-20320	25043.07	1
	TC-20980	19052.22	2
	RB-19360	15117.35	3
	TA-21385	14595.62	4
	AB-10105	14473.57	5

Result 15 x

2. Who are the bottom 5 customers?

```
345 • WITH customer_sales AS
346 (
347     SELECT
348         customer_id,
349         SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
350     FROM orders
351     GROUP BY customer_id
352 )
353
354 SELECT *
355 FROM
356 (
357     SELECT *,
358         RANK() OVER(ORDER BY total_sales ASC) AS rnk
359     FROM customer_sales
360 ) x
361 WHERE rnk <= 5;
362
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	total_sales	rnk
▶	TS-21085	4.84	1
	LD-16855	5.30	2
	CJ-11875	16.52	3
	MG-18205	16.74	4
	RS-19870	22.33	5

Result 16 x

3. Which customers made only one order?

```
363  -----
364  -- 3. Customers Who Made Only One Order
365  -----
366
367  • SELECT
368      customer_id,
369      COUNT(order_id) AS total_orders
370  FROM orders
371  GROUP BY customer_id
372  HAVING COUNT(order_id) = 1;
373
374  -----
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	total_orders
▶	JR-15700	1
	AO-10810	1
	LD-16855	1
	CJ-11875	1
	RE-19405	1

Result 17 x

4. Which customers have above-average sales?

```
377
378  • WITH customer_sales AS
379  (
380      SELECT
381          customer_id,
382          SUM(CAST(sales AS DECIMAL(10,2))) AS total_sales
383      FROM orders
384      GROUP BY customer_id
385  )
386
387  SELECT *
388  FROM customer_sales
389  WHERE total_sales >
390  (
391      SELECT AVG(total_sales)
392      FROM customer_sales
393  );
394
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	total_sales
▶	BH-11710	6255.34
	IM-15070	4930.49
	PK-19075	8646.93
	TB-21520	4737.48
	MA-17560	4299.16
	KN-20710	3127.97

Result 18 x

5. What is the highest order value per customer?

```
395  -- -----
396  -- 5. Highest Order Value Per Customer
397  -- -----
398
399  • SELECT
400      customer_id,
401      MAX(CAST(sales AS DECIMAL(10,2))) AS highest_order
402  FROM orders
403  GROUP BY customer_id;
404
```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content: 

	customer_id	highest_order
▶	CG-12520	731.94
	DV-13045	721.88
	SO-20335	957.58
	BH-11710	1706.18
	AA-10480	479.97
	IM-15070	1819.86
	HP-14815	1242.90
	PK-19075	4305.55
	AG-10270	1336.44
	ZD-21925	823.96
	KB-16585	1013.83

Result 19 ×